Assuming that at a given iteration, when temperature has the value T, the algorithm attempts to transition from current point P_0 to a point P, then we can express this probability function $A(P_0, P, T)$ as:

$$A(P, P_0, T) = \begin{cases} e^{-\frac{C(P_0)-C(P)}{k \cdot T}} & \text{if } C(P) \geq C(P_0) \\ 1 & \text{otherwise} \end{cases}$$

where $C(P)$ is the cost of solution P (similarly for P_0), and k is a constant that must be calibrated on the initial temperature and max delta in the cost function, so that the probability of acceptance of a positive delta is close to 1 for any two states, in the initial phase of the algorithm (when system's temperature is close to the initial temperature).

Typically, the probability of transitioning to a lower-energy state is set to 1, so that such a transition is always allowed, in any phase of the simulation.

But, of course, this is not the only way to define this function; it is, however, a typical definition for the acceptance probability, as it stems directly from the metallurgic analogous: this formula is directly inspired by the Boltzmann distribution, that measures the probability that a system is in a state with a certain energy and temperature[8]. For simulated annealing, instead that an absolute value for the energy we consider a variation from a lower to a higher energy state.

Now, what's the effect of this probability distribution on the single step of the algorithm? Let's consider the case where